MULTIPURPOSE SEED SIEVING MACHINE

❖ ABSTRACT

The Automatic Horizontal Sieving Machine is designed to provide an efficient, low-cost, and user-friendly solution for separating particles of different sizes in construction and agricultural applications. Traditional sieving methods are labor-intensive, time-consuming, and often produce inconsistent results, while existing automated machines are expensive and unsuitable for small-scale users. This project introduces a compact and portable sieving system that operates on a slider-crank mechanism to convert rotary motion into controlled reciprocating motion, enabling continuous and uniform sieving. The machine effectively separates materials such as sand, grains, and other granular substances by allowing smaller particles to pass through the mesh while retaining larger ones. By automating the sieving process, the system reduces manual effort, minimizes processing time, and improves accuracy and productivity. Its simple mechanical design, low maintenance requirement, and dual power capability make it suitable for construction sites, rural areas, and domestic use, offering a practical and economical alternative to conventional sieving methods.

❖ INTRODUCTION

In today's fast-paced world, efficiency, speed, and accuracy are essential in every field, especially in construction and agriculture where material handling plays a crucial role. Traditional sieving methods used for separating sand, grains, and other materials are largely manual, time-consuming, and labor-intensive. These conventional practices not only reduce productivity but also lead to inconsistent output and increased physical strain on workers. With the growing demand for quality materials and timely project completion, there is a need for modern, automated solutions that can simplify operations and enhance overall efficiency, particularly for small-scale industries and rural users.

To address these challenges, the Automatic Horizontal Sieving Machine is developed as a compact, cost-effective, and efficient solution for material separation. The machine operates on the principle of converting rotary motion into reciprocating motion using a slider-crank mechanism, enabling continuous and uniform sieving. It is designed with simplicity, portability, and ease of operation in mind, making it suitable for use in construction sites, agricultural fields, and domestic applications. By automating the sieving process, the machine reduces manual effort, saves time, and ensures consistent results, thereby

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improving productivity and providing a practical alternative to traditional sieving methods.

❖ PROJECT OBJECTIVES

• To increase the productivity and efficiency of sieving operations for sand, grains, and other materials.
• To reduce manual labor and human effort involved in traditional sieving methods.
• To minimize the time required for filtering and separating materials.
• To develop a compact, portable, and easy-to-transport sieving machine.
• To achieve uniform and consistent particle separation through controlled reciprocating motion.
• To design a user-friendly system that can be operated without requiring skilled labor.

❖ MOTIVATION OF THE PROJECT

• The increasing demand for faster and more efficient material handling in construction and agriculture inspired the need for an automated sieving solution.
• Traditional manual sieving methods are time-consuming, labor-intensive, and produce inconsistent results, motivating the development of a more reliable system.
• Workers involved in manual sieving are exposed to dust and physical strain, creating a need for a safer and healthier alternative.
• Existing automated sieving machines are expensive and complex, making them inaccessible to small-scale industries and rural users.
• The need to reduce dependency on skilled labor and simplify operations encouraged the design of a user-friendly machine.

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❖ PROBLEM OF THE STATEMENT

The existing sieving methods used in construction, agriculture, and small-scale industries are largely inefficient, time-consuming, and labor-intensive. Manual sieving requires significant physical effort, provides inconsistent particle separation, and becomes impractical when handling large quantities of materials such as sand and grains. Additionally, commercially available sieving machines are often expensive, complex to operate, and not suitable for small contractors, rural users, or domestic applications due to their high cost and maintenance requirements. The lack of portable, affordable, and efficient sieving solutions leads to reduced productivity, delays in work completion, and compromised material quality. Therefore, there is a need to develop a compact, low-cost, and automated sieving machine that can reduce manual effort, improve efficiency, and provide consistent results while being easy to operate and suitable for a wide range of users.

❖ WORKING OF THE PROJECT

The Automatic Horizontal Sieving Machine works on the principle of converting rotary motion into reciprocating motion to perform efficient sieving. When the machine is powered on, the motor generates rotary motion, which is transmitted through a pulley mechanism to the crank system. The crank and slider mechanism then converts this rotary motion into a horizontal reciprocating motion of the sieving box. The sieve box is placed on a guided rail track, ensuring smooth and controlled back-and-forth movement. When sand, grains, or other materials are placed on the metallic mesh, the continuous reciprocating motion causes smaller particles to pass through the mesh while larger particles remain on the surface. This results in effective and uniform separation of materials. The machine operates with minimal vibration, reduced manual effort, and consistent performance, making the sieving process faster, more efficient, and suitable for various applications.

❖ HARDWARE REQUIREMENTS

1. SLIDER
2. CASTER WHEELS
3. SUPPORTING FRAME
4. SHAFT
5. BEARINGS

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6. METALLIC MESH (SIEVE)
7. 12V BATTERY
8. SPEED CONTROLLER
9. JOHNSON MOTOR (HIGH TORQUE MOTOR)

❖ ADVANTAGES OF THE PROJECT

• Increases productivity by performing sieving operations faster than manual methods.
• Reduces human effort and minimizes physical strain on workers.
• Provides uniform and consistent particle separation with higher accuracy.
• Saves time and improves overall workflow efficiency in construction and agriculture.
• Easy to operate and does not require skilled labor.
• Portable and compact design allows easy transportation to different locations.
• Can operate on both electrical power and battery, making it suitable for remote

❖ APPLICATIONS OF THE PROJECT

- Construction Industry Used for sieving sand to obtain fine particles required for plastering, concrete mixing, and brickwork.
- Agriculture Helps in cleaning and separating grains, seeds, and spices by removing dust, stones, and impurities.
- Small-Scale Industries Suitable for material separation processes in workshops and cottage industries.
- Domestic Use Can be used in households for cleaning food items such as grains, pulses, and flour.
- Foundries Used for filtering and grading molding sand in metal casting processes.

❖ IN THIS PROJECT, THERE ARE FOUR KITS AVAILABLE

1. **Ready-made Working Project kit.** The ready-made project kit along with comprehensive documentation (over 60 pages report, 15 to 20 slides PPT, circuit diagram, program code, PCB layout and schematic, explanatory video), audio and video call support, and a one-year warranty, priced at 10000 Rs.

2.The "**For Documentation**" option provides comprehensive documentation (over 60 pages report, 15 to 20 slides PPT, circuit diagram, program code, PCB layout and schematic, explanatory video), along with audio call support, for a price of 2000 Rs.

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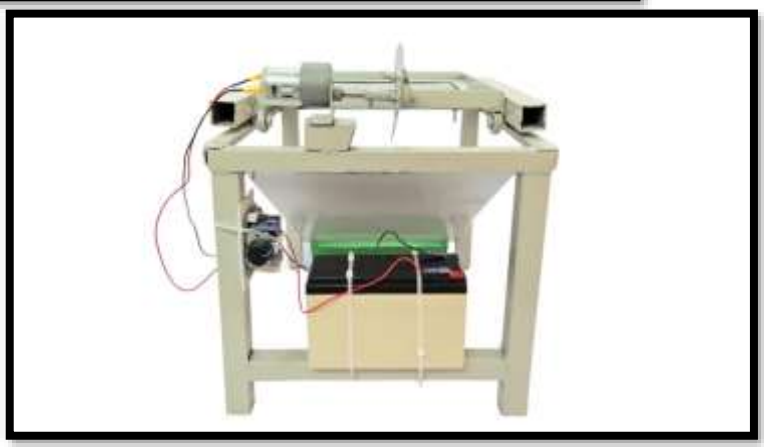
❖ FINAL OUTPUT LOOK O F THE PROJECT